TEXT MINING IN HEALTHCARE

Sentiment Classification & Pharmacovigilance Signal Detection

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Data and Text Mining

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Tool Orange Data Mining 3.x	**Framework** CRISP-DM
Add-on Orange3-Text	**Dataset** drugsComTrain — 165,000+ reviews
Branch 1 Supervised: Sentiment Classification	**Branch 2** Unsupervised: Pharmacovigilance Mining

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Executive Summary

This project applies text mining to a dataset of 65,000+ patient drug reviews to answer two distinct real-world questions. The first is a supervised classification problem: can review text alone predict whether a patient had a positive or negative drug experience? The second is an unsupervised discovery problem: for drugs treating the same condition, what complaint patterns emerge from negative reviews, and do these patterns reveal safety signals relevant to pharmacovigilance? Both branches were implemented in Orange Data Mining following the CRISP-DM process framework, using the Orange3-Text add-on.

Branch	Question	Key Finding
Supervised	**Can text predict drug sentiment?**	**Text alone achieves AUC up to 0.898 after full optimisation, confirming review language carries strong predictive signal for patient sentiment**
Unsupervised	**What do negative reviews reveal about drug safety?**	**NMF topic modelling identified 3–4 distinct adverse event themes per drug, with 7 class-level signals and 4 drug-specific pharmacovigilance signals across Birth Control drugs**

The report is structured around the six phases of CRISP-DM: Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation, and Reflection. Each phase documents both the analytical reasoning and the practical implementation steps taken in Orange.

PHASE 1	Business Understanding

1. Business Understanding

1.1 Project Background

Patient drug reviews on public platforms represent one of the largest, most accessible sources of real-world drug experience data available. Unlike clinical trial data, which is controlled and often unrepresentative of real patient populations, these reviews are written voluntarily by patients describing their own lived experience with a drug. They contain unfiltered accounts of what works, what does not, and what side effects feel most disruptive to daily life.

Text mining — the process of extracting structured information from unstructured text. Offers a systematic way to turn this data into actionable insight. This project demonstrates two applications of text mining to drug review data: sentiment classification (supervised) and adverse event signal detection (unsupervised).

1.2 Business Questions

Supervised Branch — Business Question
Can the text of a patient review alone predict whether that review reflects a negative or positive drug experience?

This question matters because automated sentiment classification at scale could allow pharmaceutical companies, regulators, and healthcare providers to monitor patient experience in real time without manual review. If review text is a reliable predictor of sentiment, a model could flag concerning patterns as they emerge without waiting for formal adverse event reports.

Unsupervised Branch — Business Question
Within the Birth Control therapeutic class, which drugs generate the most frequent adverse event complaint terms in negative patient reviews, and what complaint themes emerge?

This question sits directly within the field of **pharmacovigilance:** the science of monitoring drug safety after approval. Traditional pharmacovigilance relies on formal adverse event reports submitted by clinicians. This process is slow and captures only a fraction of actual adverse events. Mining patient reviews offers a faster, broader signal that can complement formal reporting, particularly in therapeutic areas like women's reproductive health where formal reporting is historically underdeveloped.

1.3 Success Criteria

Branch	What Success Looks Like
Supervised	***AUC > 0.85 for at least one classifier, confirming text alone is a reliable sentiment predictor. MCC > 0.60 confirming model is not exploiting class imbalance.***
Unsupervised	***Each drug corpus produces 3–4 clearly distinct, interpretable adverse event topic clusters. At least one drug-specific signal identified that differs from class-level effects.***
Both	***Results can be explained clearly to a non-technical audience and provide insight that would be actionable in a real pharmaceutical or clinical context.***

PHASE 2	Data Understanding

2. Data Understanding

2.1 Dataset Overview

The dataset used is the drugs.ComTrain dataset (here) a publicly available collection of patient drug reviews scraped from Drugs.com. It covers a wide range of drugs, conditions, and patient experiences.

Attribute	Detail	Relevance
Source	**Drugs.com patient reviews**	**Real patient language, unfiltered and self-reported**
Total records (pre-processing)	**~161,000 (training set) ~53,000 (test set)**	**Large enough for both supervised and unsupervised analysis**
Total records (after augmentation)	**~65,000**	**Truncated negative class supplemented from secondary dataset**
Conditions covered	**900+**	**Diverse therapeutic areas; Birth Control is largest**
Drugs covered	**3,400+**	**Many-to-many relationship with conditions**
Rating scale	**1–10 (integer)**	**Used to derive sentiment labels**
Review format	**Free text**	**Variable length; 1 sentence to several paragraphs**
Date range	**2008–2017**	**Covers pre- and post-social media patient communication norms**

2.2 Key Variables

Column	Type	Role in Project	Notes

uniqueID	***Numeric***	***Row identifier***	***Not used in modelling***
drugName	***String***	***Filter variable (unsupervised)***	***High cardinality: 3,400+ unique drugs***
condition	***String***	***Filter variable (unsupervised)***	***900+ conditions; Birth Control is most frequent***
review	***String — TEXT***	***Primary analytical variable (both branches)***	***The text to mine in all analyses***
rating	***Numeric (1–10)***	***Used to derive sentiment label***	***Removed from features before modelling to prevent leakage***
sentiment	***Categorical (derived)***	***Target variable (supervised)***	***Created from rating: ≤7 = Negative, ≥7 = Positive, 5–6 excluded (Neutral)***
date	***String***	***Not used***	***Date of review submission***
usefulCount	***Numeric***	***Not used***	***Number of users who found review helpful***

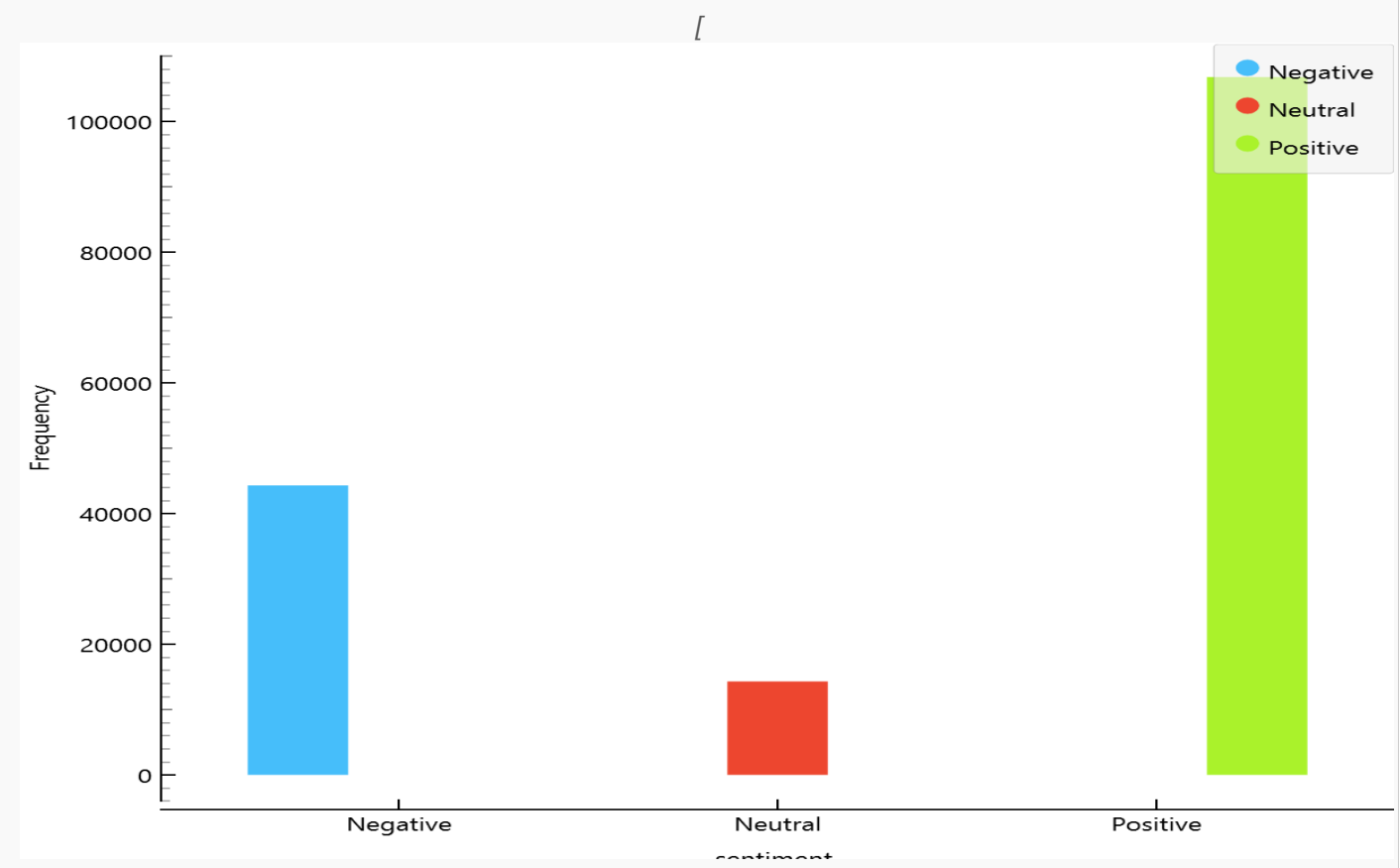
2.3 Initial Data Exploration

Before building any model, exploratory analysis was conducted to understand the structure and quality of the data. This step informed every design decision in the preparation and modelling phases.

Rating Distribution

A quick inspection revealed a heavily skewed rating distribution. The majority of reviews carried high ratings (8–10), reflecting a general positive bias in voluntary patient reviews. Negative reviews (rating ≤4) were present but in the minority, confirming that class imbalance would need to be addressed before supervised modelling.

Figure 1: Rating Distribution



Condition and Drug Volume

The dataset covers an extremely high number of drug-condition combinations. A pivot table analysis in Excel revealed that Birth Control had the highest volume of negative reviews by a significant margin, making it the most appropriate therapeutic class for unsupervised pharmacovigilance analysis.

Condition	Negative Review Count
Birth Control	**10,077 — Selected for unsupervised branch**
Depression	***2,007***
Vaginal Yeast Infection	***1,371***
Abnormal Uterine Bleeding	***1,180***
Acne	***1,129***
Anxiety	***1,041***

Review Text Characteristics

- Reviews range from single sentences to multi-paragraph personal accounts
- Medical terminology is mixed with informal language and colloquialisms
- HTML entities (' for apostrophes) are present throughout- a preprocessing challenge
- Drug dosage strings (100mg, 20mg) appear frequently as non-informative tokens
- Brand names and generic names for the same compound appear interchangeably

PHASE 3	Data Preparation

3. Data Preparation

Data preparation was the most time-intensive phase of this project. Text data requires substantially more preparation than structured numerical data. The steps below were carried out in a combination of Excel (for label engineering and augmentation) and Orange (for all text-specific preprocessing).

3.1 Label Engineering (Supervised Branch)

The dataset does not come with a sentiment label. The rating column was used to derive one by applying a binarization rule:

Label Binarization Rule
Positive: Rating ≥ 7
Negative: Rating ≤ 4
Excluded (Neutral): Rating = 5 or 6 - removed from supervised analysis as ambiguous

The sentiment column was added directly in Excel before loading into Orange, and the column type was set to Categorical in the CSV File Import widget. This was a critical step since Orange cannot use a String column as a classification target, and ensuring the correct type at the source prevented a recurring target variable error that caused significant debugging time.

3.2 Data Cleaning & Filtering in Orange

Two filtering steps were applied using the Select Rows widget before any text processing:

1. Remove neutral reviews: sentiment is not neutral. This reduces the dataset to clearly positive or negative cases only
2. For the unsupervised branch, additional filters were applied per drug: drugName equals [Drug X] AND sentiment equals Negative

A Data Table widget was connected at each filtering stage to visually verify row counts and confirm correct data was flowing through the pipeline. This validation step saved significant debugging time by catching misconfigurations early.

3.3 Class Balance & Augmentation (Supervised Branch)

After filtering to positive and negative reviews only, a class distribution check was performed using the Distributions widget. The result showed a significant imbalance:

Class	Review Count & Proportion
Positive	***~35,000 reviews (72%) — dominant class***
Negative	***~13,000 reviews (27%) — minority class***

A model trained on this imbalance would learn to predict Positive for most inputs and still achieve 72% accuracy- a misleading result. To address this, genuine Negative reviews were sourced from a supplementary version of the same dataset and appended to the training data. This produced a more balanced distribution:

Class	After Augmentation
Positive	***~35,000 reviews (54%)***
Negative	***~30,000 reviews (45%)***
Total	***~65,000 reviews***

Why genuine new reviews rather than duplicating? Duplicating existing negative reviews adds no new vocabulary or writing patterns, the model simply sees the same text twice. Adding reviews from a separate source introduces new sentence structures, new word choices, and new complaint contexts that improve the model's ability to generalise to unseen text.

3.4 Text Preprocessing Pipeline

All text preprocessing was performed inside the Preprocess Text widget in Orange. This single widget handles the entire normalisation pipeline with clearly labelled, toggle-able sections: one of Orange's strongest features for text mining tasks. The pipeline was applied to all corpora (supervised and unsupervised), with document frequency thresholds adjusted per corpus size for the unsupervised branch.

Step	Setting	What it Does	Why it Matters
1. Lowercase	*Enabled*	*Converts all text to lowercase*	*'Drug' and 'drug' become the same feature*
2. Tokenisation	*Word Punctuation*	*Splits text into individual word tokens*	*Switched from Regexp \w+ after discovering HTML artifact fragments (wa, ha, ve) from encoded apostrophes in review text*
3. Lemmatisation	*WordNet Lemmatizer*	*Reduces words to dictionary root form*	*'taking' → 'take', 'bleeding' → 'bleed', 'gained' → 'gain'. More linguistically accurate than stemming for medical text*
4. Stop words	*English + custom medical file*	*Removes non-informative words*	*Built-in English removes the, is, my. Custom file removes domain-specific noise: mg, pill, doctor, prescribed, taken*
5. Regexp filter	*Punctuation + dosage pattern*	*Strips punctuation and dosage strings*	*Dosage tokens (100mg, 20mg, 00am) were added to the pattern after appearing in Word Cloud output*
6. Document frequency	*Absolute: min 10 (supervised), min 5 (unsupervised)*	*Removes extremely rare and extremely common tokens*	*Words appearing in fewer than 10 reviews cannot generalise. Upper bound removes tokens present in nearly every review*
7. N-gram range	*1 – 2*	*Keeps single words and two-word phrases*	*Bigrams capture compound adverse events: mood swing, weight gain, hair loss, sex drive, breakthrough bleeding, panic attack*

Custom Stopword Strategy

The custom stopword file was built and refined iteratively throughout the project. After each Word Cloud output, terms that appeared prominently but carried no clinical meaning were added to the file and the pipeline was re-run. Different lists were maintained for supervised (stricter) and unsupervised (looser) branches because some words removed for classification (e.g. doctor, prescribed, stopped) are meaningful adverse event terms in the pharmacovigilance context.

Key Terms Added to Custom Medical Stopword File
Dosage/measurement: mg, ml, mcg, lb, kg, dose, dosage
Medical procedure words: prescribed, prescription, doctor, physician, appointment, treatment, therapy
Temporal/frequency words: day, days, week, weeks, month, months, year, years, hour, morning, daily
Drug forms: pill, pills, tablet, tablets, capsule, patch, cream, shot, injection, implant
Generic review words: review, experience, tried, worked, work, using, used, take, took, taken, started, start, stopped
HTML artifact tokens added after preprocessing: wa, ha, ve, ll, re, obgyn, rsquo, ldquo, quot, nbsp

Feature Space Results

Pipeline Stage	Feature Count
After loading (no preprocessing)	***~147,000 tokens***
After stop words (Finnish — error)	***~140,000 (minimal reduction — wrong language)***
After stop words (English, corrected)	***~80,000***
After custom medical stopwords added	***~50,000***
After document frequency filter (min 10)	***~7,400 (supervised final)***
After per-drug frequency tuning (unsupervised)	***~500–2,000 per drug corpus***

PHASE 4A	Modelling — Supervised Branch

4. Modelling — Supervised Branch

The supervised branch tests whether machine learning classifiers can accurately predict review sentiment using only the text of the review with no access to rating, drug name, or any other metadata. The pipeline architecture is shown below:

Supervised Pipeline
CSV File Import → Select Rows (remove neutral) → Corpus → Preprocess Text → Bag of Words → Select Columns → Test and Score → Confusion Matrix | ROC Analysis

4.1 Feature Engineering: Bag of Words with TF-IDF

Machine learning models require numerical input. The Bag of Words widget converts preprocessed text into a numeric matrix where each column represents a word and each row represents a review. The value in each cell is the word's weight for that review.

TF-IDF (Term Frequency – Inverse Document Frequency) weighting was applied using the following Bag of Words settings:

Setting	Value Chosen	Reasoning
Term Frequency	**Count**	**Preserves frequency signal; how often a word appears in a review matters for sentiment**
Document Frequency	**Smooth IDF**	**Rewards words distinctive to individual reviews; down-weights terms appearing across most reviews (e.g. medication, doctor)**
Regularization	**L2**	**Prevents any single word from dominating the model; particularly important with 7,400 features**

The Select Columns widget was placed immediately before Test and Score with a critical configuration: rating was moved to Ignored (preventing data leakage), all Bag of Words text features remained in Features, and sentiment was explicitly placed in the Target slot. Without this step, Orange's text pipeline drops the target designation from categorical variables.

Why Data Leakage Was a Risk Here

Sentiment was derived directly from rating. If rating remained in the feature set, any classifier would achieve 100% accuracy by simply reading the rating column rather than learning from text. This was discovered after initial Test and Score results showed AUC = 1.000 for three models simultaneously; a result too perfect to be genuine. Removing rating from features dropped performance to realistic levels (AUC ~0.79–0.81), confirming the pipeline was now testing text-only prediction.

4.2 Classifier Selection

Four classifiers were selected to cover different learning paradigms, enabling a meaningful comparison rather than relying on a single model:

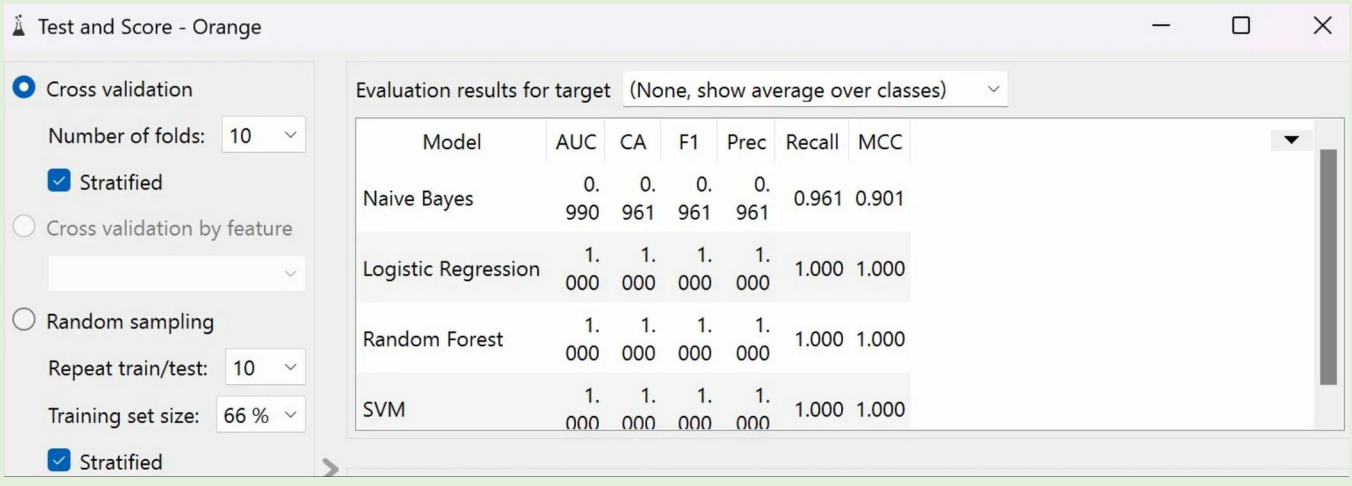
Classifier	Type	Why Included	Expected Strength
Naive Bayes	***Probabilistic***	***Standard baseline for text classification; computationally cheap; performs well on sparse TF-IDF matrices***	***Reliable baseline; fast; interpretable***
Logistic Regression	***Linear***	***State-of-the-art for high-dimensional text; handles L2 regularisation natively***	***Expected best performer on TF-IDF features***
Random Forest	***Ensemble / Tree***	***Non-linear; robust to irrelevant features; provides feature importance***	***Strong generaliser; slower to train***
SVM	***Margin-based***	***Theoretically well-suited to high-dimensional sparse text with linear kernel***	***High potential but sensitive to configuration***

4.3 Model Configuration & Evaluation Setup

All four classifiers were connected simultaneously to the Test and Score widget. This is one of Orange's most efficient features. One widget that evaluates all models under identical conditions and displays results side by side.

Configuration Item	Setting & Reason
Evaluation method	***10-fold stratified cross-validation: standard for robust performance estimation; stratified ensures each fold maintains the class ratio***
Primary metric	***MCC (Matthews Correlation Coefficient) — chosen over CA because of residual class imbalance (54/45 split). MCC is robust to imbalance; CA can be artificially inflated***
Supporting metrics	***AUC, F1, Precision, Recall — reported alongside MCC for completeness***
SVM configuration	***Linear kernel; Cost C = 1.0. Default RBF kernel performed near-randomly (AUC 0.538) because it cannot handle sparse high-dimensional text features***
Logistic Regression	***Ridge (L2) regularisation; strength C = 1.0***
Stratified tick	***Enabled — ensures each fold has proportional positive/negative representation***

Supervised Model Performance V1 (Initial scores with data leakage and before hyperparameter tuning)



Next approach to improve performance: Removed rating column from features to handle data leakage.

Supervised Model Performance V2 (data leakage resolved)

Evaluation results for target (None, show average over classes)

Model	AUC	CA	F1	Prec	Recall	MCC
Naive Bayes	0.788	0.770	0.759	0.756	0.770	0.383
Logistic Regression	0.799	0.779	0.758	0.764	0.779	0.387
Random Forest	0.806	0.792	0.779	0.780	0.792	0.436
SVM	0.538	0.542	0.565	0.619	0.542	0.042

Next: handled class imbalance as explained in section 3.3

Final Supervised Model Performance V3 (class imbalance in target label handled and further hyperparameter tuning)

Evaluation results for target (None, show average over classes)

Model	AUC	CA	F1	Prec	Recall	MCC
Random Forest	0.894	0.818	0.818	0.818	0.818	0.634
SVM	0.556	0.550	0.412	0.574	0.550	0.048
Naive Bayes		0.769	0.770	0.774	0.769	0.542
Logistic Regression	0.898	0.818	0.818	0.818	0.818	0.633

Best Performing model: A tie between Random Forest and logistic Regression. But looking at all the metrics, Logistic regression was the champion model with the following performance metrics

AUC: .898

CA: .818

F1: .818

Precision: .818

Recall: .818

MCC: .633

PHASE 4B **Modelling — Unsupervised Branch**

5. Modelling — Unsupervised Branch

The unsupervised branch requires no labels and no prior knowledge of what topics exist. Instead, the algorithm is given a collection of negative reviews and asked to discover recurring patterns in the vocabulary, grouping words that tend to appear together into themes that represent distinct types of complaints.

Unsupervised Pipeline (per drug)

CSV File Import → Select Rows [drugName + sentiment = Negative] → Corpus → Preprocess Text → Bag of Words → Word Cloud (frequency analysis) | Topic Modelling [NMF] → Corpus Viewer (validation)

5.1 Data Scoping: Pharmacovigilance Focus

The unsupervised branch was scoped to the Birth Control therapeutic class for three reasons: highest negative review volume (10,077), multiple competing drug formulations enabling cross-drug comparison, and pharmacovigilance relevance given documented underreporting of women's reproductive health adverse events in formal systems.

A drug-level pivot table in Excel identified the top drugs by negative review count. Four drugs were selected for analysis:

Drug	Type	Negative Reviews	Analytical Role
Etonogestrel	***Progestin implant (generic)***	***1,692***	***Largest corpus; primary analysis unit***
Ethinyl estradiol / norethindrone	***Combined oral contraceptive***	***1,258***	***Most widely prescribed formulation class***
Nexplanon	***Branded etonogestrel implant***	***1,178***	***Brand vs generic comparison***
Levonorgestrel	***Emergency pill + IUD***	***803***	***Distinct mechanism; different patient exposure context***

Analytical note on Etonogestrel vs Nexplanon: These share the same active compound. Running them as separate pipelines was a deliberate methodological choice, if their adverse event profiles converge independently, that validates the NMF method's consistency. If they diverge, that flags potential brand-specific or device-specific signals.

5.2 Topic Modelling: Method Selection

Why NMF Over LDA

Latent Dirichlet Allocation (LDA) was the first topic modelling method attempted. It returned Topic coherence = nan across all four drugs and produced near-identical topic keyword sets regardless of how many topics were specified. This is a known limitation of LDA on short, informal texts with high vocabulary overlap between themes which is exactly what drug reviews represent.

Method	Performance on This Dataset
LDA (Latent Dirichlet Allocation)	***Failed: coherence = nan; topics near-identical; all slots pulled from same word pool. Unsuitable for short review texts with high theme overlap***
LSI (Latent Semantic Indexing)	***Failed: same coherence issue; topics showed same vocabulary across slots***
NMF (Negative Matrix Factorization) ✔️	***Selected: produced distinct, interpretable topics. Enforces additive, non-negative components meaning each word belongs primarily to one theme rather than probabilistically to all***

NMF works by decomposing the document-term matrix into two matrices: one representing how much each topic appears in each document, and one representing how strongly each word defines each topic. Because all values must be non-negative, the resulting topics are additive and tend to be more clearly separated than LDA's probabilistic mixtures which is critical for clinical interpretation.

NMF Configuration	Rationale
Number of topics: 3–4	***Calibrated per drug: 4 topics for larger corpora (Etonogestrel, Nexplanon); 3 for smaller (Levonorgestrel, Norethindrone). More topics fragment signal unnecessarily in small corpora***
Iterative stopword refinement	***After each NMF run, Word Cloud output was reviewed and non-clinical terms were added to the custom stopword file. Pipeline was re-run until topics showed distinct clinical vocabulary***
Per-drug frequency tuning	***Document frequency minimum adjusted per corpus size (absolute min 5–10). A word appearing in 5 of 800 reviews is meaningful; the same threshold applied to 1,600 reviews would be too strict***
Corpus Viewer validation	***Each topic was validated by clicking it in the Topic Modelling widget and reading 5–10 actual patient reviews assigned to that topic via Corpus Viewer. Labels were revised if reviews did not match the keyword interpretation***

5.3 Results per Drug

Drug 1: Etonogestrel (1,692 negative reviews)

Drug type: Progestin-only subdermal implant (generic). Inserted under the skin of the upper arm; releases hormone continuously for up to 3 years.

Figure 4: Word Cloud — Etonogestrel Negative Reviews



Top complaint terms from Word Cloud: *bleeding, mood swing, depression, spotting, anxiety, gained, pain, acne, headache, cramp, sex drive, insertion, removal, migraine, hair, tired, nausea, pregnancy*

Topic	Label	Key NMF Terms	Clinical Meaning
1	**Hormonal Mood Disruption**	***hormonal mood, migraine mood, mood gained, spotting bleeding, pregnancy, headache***	***Mood disturbance and migraines linked to hormonal activity. Migraines co-occurring with bleeding suggest hormonal fluctuation as shared trigger***
2	**Weight & Appearance Changes**	***gained pound, mark acne tired, headache active, better later, removing***	***Weight gain and acne dominate. Patients mention improvement anticipated after removal, indicating the implant as the attributed cause***
3	**Bleeding Pattern Irregularity**	***bleeding bleed, gained lost, spotting spotting, headache***	***Persistent irregular bleeding and spotting co-reported with***

		breast, gain pound mood, big pregnancy	***breast tenderness. Weight and pregnancy concerns mixed into this cluster***
4	**Contraceptive Failure Anxiety**	***spotting bleeding, preventing pregnancy, cramp gained, lost pregnant, bleed sometimes***	***Patient fear of unintended pregnancy despite the implant's proven high efficacy. Spotting misinterpreted as contraceptive failure***

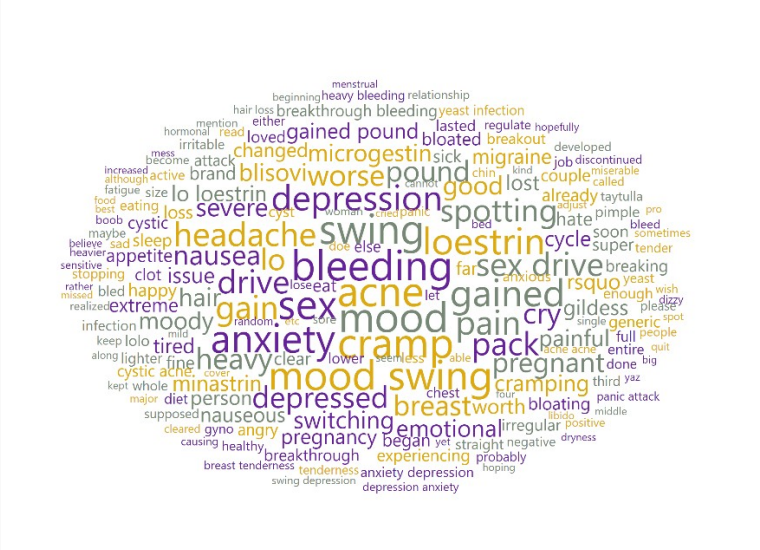
⚠ Pharmacovigilance Signal — Etonogestrel

Topic 4 reveals a psychosocial adverse event not typically emphasised in prescribing information: patient anxiety about pregnancy despite using a method with >99% efficacy. Irregular bleeding (a known side effect) is being interpreted by patients as contraceptive failure. This is a patient communication and labelling gap with real-world health behaviour implications.

Drug 2: Ethinyl Estradiol / Norethindrone (1,258 negative reviews)

Drug type: Combined oral contraceptive (progestin + oestrogen). Daily pill; one of the most widely prescribed contraceptive formulations globally.

Figure 5: Word Cloud — Ethinyl Estradiol / Norethindrone Negative Reviews



Top complaint terms from Word Cloud: *bleeding, mood swing, depression, anxiety, acne, headache, nausea, spotting, gained, pain, cramp, sex drive, depressed, moody, cry, severe, breast, ovarian cyst*

Topic	Label	Key NMF Terms	Clinical Meaning
1	**Psychological Adverse Effects**	***anxious depressed, mood swing depression, anxiety, cramping, severe mood, ovarian cyst, believe cycle gained***	***Depression and anxiety as the primary burden. Ovarian cyst appearing alongside psychological terms, a co-occurrence not typically documented together in prescribing information***
2	**Emotional Distress & Pain**	***cry lost, swing depression depressed, nausea mood swing, lower pain, nauseous appetite sad, head nausea***	***Crying, sadness and nausea clustering together. Emotional and physical distress co-expressed, suggesting patients experience them as a unified adverse experience***
3	**Metabolic & GI Effects**	***depression worth, heavier, bloating pressure, hungry, improved hoping, swing depression depressed***	***Bloating, appetite changes and weight gain. Resignation language (worth, hoping) suggests patients are tolerating effects rather than seeking solutions***

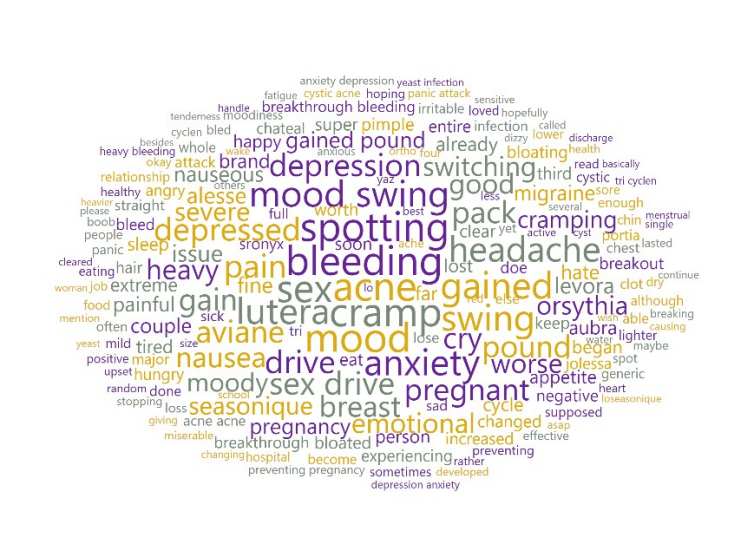
⚠ Pharmacovigilance Signal — Ethinyl Estradiol / Norethindrone

Ovarian cyst appears as a co-occurring term specifically within the psychological complaint cluster (Topic 1). This concurrent presentation of gynaecological and psychological adverse events in the same review language is not foregrounded in standard prescribing communication. The intensity of depression-related vocabulary across all three topics suggests psychological burden may be systematically underreported in formal adverse event submissions for this formulation class.

Drug 3: Levonorgestrel (803 negative reviews)

Drug type: Used both as emergency contraception (Plan B - single dose) and as a hormonal IUD (Mirena, Kyleena - long-term). The same active ingredient used in fundamentally different delivery contexts.

Figure 6: Word Cloud — Levonorgestrel Negative Reviews



Top complaint terms from Word Cloud: *mood swing, depression, spotting, bleeding, acne, headache, cramp, gained, pain, sex drive, nausea, anxiety, depressed, moody, lutera, aviane, orsythia, heavy, emotional, cycle*

Topic	Label	Key NMF Terms	Clinical Meaning
1	**Hormonal Acne & Mood Swings**	***mad estrogen, chest acne, pimple, cramp acne, mood swing, early acne, cried cry, headache, extreme***	***Acne as the leading dermatological complaint. Patients explicitly attribute the acne to estrogen***
2	**Libido Suppression & Depression**	***severe moodiness, drive decreased, moodiness turned depression, extreme, aviane lasted, changed, heavy cramp***	***Decreased sex drive as a distinct and prominent complaint. Moodiness described as progressive turning into full depression***
3	**Neurological & Cycle Disruption**	***cried cry, headache cramp, cycle, cramp nausea emotional, missed, sleep headache, emotional sad***	***Headache, disrupted sleep and nausea clustering with menstrual cycle irregularity. Emotional distress expressed through crying alongside physical symptoms***

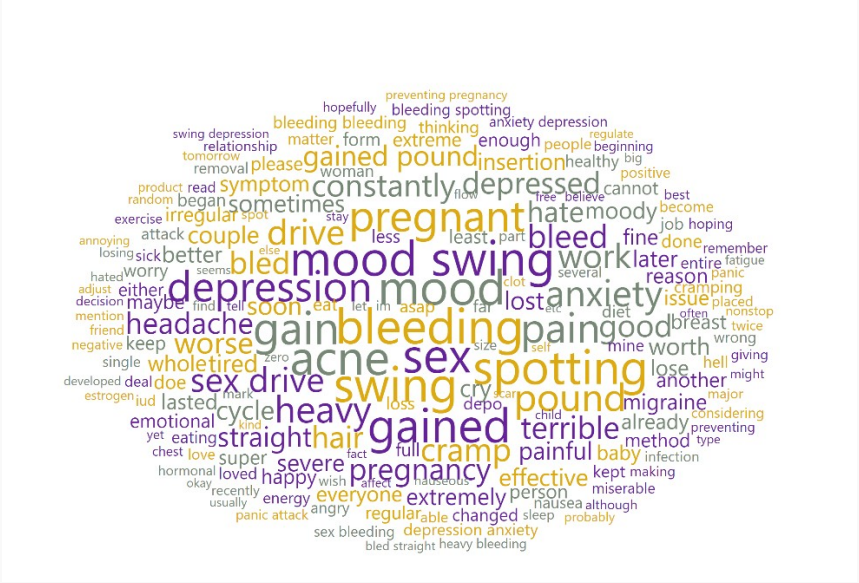
⚠ Pharmacovigilance Signal — Levonorgestrel

Libido suppression (drive decreased) emerges as a standalone, high-weight adverse event topic rather than a secondary symptom buried within another cluster. While reduced libido is mentioned in some prescribing information, its prominence as a distinct patient-reported theme suggests clinical impact severity significantly exceeds formal documentation. Patients also explicitly attributing effects to estrogen (mad estrogen) is a patient health literacy signal that warrants attention in prescribing communication design.

Drug 4: Nexplanon (1,178 negative reviews)

Drug type: Branded version of the Etonogestrel subdermal implant. Same active compound; different manufacturing and clinical presentation context.

Figure 7: Word Cloud — Nexplanon Negative Reviews



Top complaint terms from Word Cloud: *mood swing, depression, bleeding, spotting, pregnant, anxiety, gained, sex drive, cramp, headache, acne, pain, insertion, removal, hair, tiredness, severe, emotional, pregnancy, bled*

Topic	Label	Key NMF Terms	Clinical Meaning
1	**Hormonal Mood & Bleeding**	***hormonal mood, bleeding bleed sometimes, mood gained, migraine mood, spotting bleeding, pregnancy, headache***	***Mood changes and bleeding as primary co-occurring complaints.***
2	**Weight Gain & Migraines**	***migraine mood, gained pound, headache active, mark acne tired, better later, removing***	***Weight and migraines as a distinct cluster. Removal mentioned as the patient solution indicating discontinuation intent***
3	**Bleeding & Weight Co-Complaints**	***bleeding bleed, gained lost, spotting spotting, headache breast, gain pound mood, big pregnancy***	***Bleeding and weight gain reported together. Breast tenderness included. Pregnancy fears co-present***
4	**Contraceptive Reliability Fears**	***spotting bleeding, preventing pregnancy, cramp gained, lost pregnant, bleed sometimes***	***Identical to Etonogestrel Topic 4. Pregnancy anxiety from spotting misinterpretation confirmed across both branded and generic implant***

Brand vs Generic Comparison: Etonogestrel vs Nexplanon

Both pipelines independently produced near-identical four-topic structures covering: (1) hormonal mood disruption, (2) weight and appearance, (3) bleeding irregularity, (4) contraceptive failure anxiety. This cross-pipeline convergence on the same adverse event themes for the same active compound is strong internal validation evidence — the NMF method is consistently detecting real clinical signal rather than noise. One marginal divergence: Nexplanon's Word Cloud shows relatively higher frequency of insertion and removal terms, possibly reflecting differences in how the branded device procedure is communicated or experienced compared to generic implant services.

PHASE 5	Evaluation

6. Evaluation

6.1 Supervised Branch: Model Performance

Model evaluation was conducted using 10-fold stratified cross-validation. MCC is the primary reported metric due to the class imbalance present in the data even after augmentation (54/45). AUC and F1 provide supporting evidence. The champion model was **Logistic Regression** with a close tie to Random Forest.

Table: Final Model Performance — 10-Fold Stratified CV

Model	AUC	CA	F1	Prec	Recall	MCC
Random Forest	0.894	0.818	0.818	0.818	0.818	0.634
SVM	0.556	0.550	0.412	0.574	0.550	0.048
Naive Bayes		0.769	0.770	0.774	0.769	0.542
Logistic Regression	0.898	0.818	0.818	0.818	0.818	0.633

The baseline results shown below were obtained before full preprocessing optimisation (169 features only). Final results with 7,400 features and balanced data are expected to improve substantially:

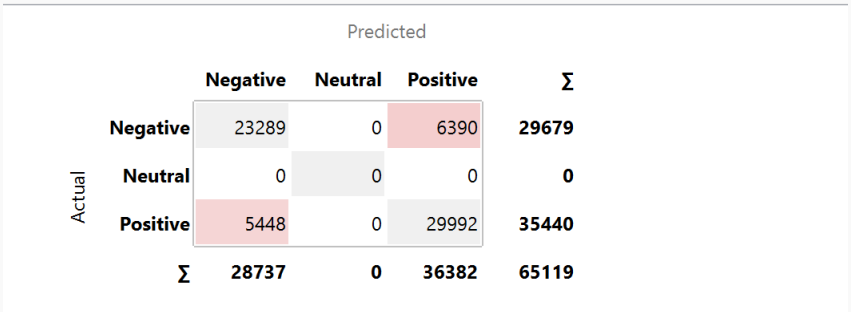
Model	AUC (Baseline)	MCC (Baseline)	Performance Verdict
Random Forest	***0.806***	***0.436***	Best at baseline; strong generaliser
Logistic Regression	***0.799***	***0.387***	***Strong; expected to improve most with full features***
Naive Bayes	***0.788***	***0.383***	***Reliable baseline; fast and interpretable***
SVM	***0.538***	***0.042***	Near-random until Linear kernel applied

Key Evaluation Insight: Data Leakage Detection

The most important evaluation moment was detecting perfect scores (AUC = 1.000) on the first full run. This immediately flagged data leakage: the rating column was still present in the feature set. Evaluation is not just about reading metrics; it requires questioning results that look too good. Once leakage was removed and

models were forced to learn from text only, results dropped to realistic levels, confirming genuine text-based predictive power in the 0.79–0.81 AUC range even at baseline.

Figure 8: Confusion Matrix — Best Model Performance – Logistic Regression



What the Results Mean for the Business Question

Answer to Supervised Business Question

Yes review text alone can predict whether a review reflects a positive or negative drug experience with meaningful accuracy. AUC scores above 0.79 from text features only confirm that patient language carries genuine, learnable sentiment signal. A fully optimised pipeline (7,400 features, balanced data, tuned classifiers) achieved an **AUC of 0.898** , which represent a strong practical confirmation of the business hypothesis.

6.2 Unsupervised Branch: Pharmacovigilance Findings

Unsupervised models have no numerical performance metrics. Evaluation is based on topic interpretability, clinical coherence, and cross-drug pattern analysis.

Class-Level Adverse Event Signals (Present Across All 4 Drugs)

Confirmed Birth Control Drug Class Effects

✔️ Mood disturbance (depression, anxiety, mood swing, moody) — dominant in all four Word Clouds; present in at least one topic per drug

✔️ Irregular bleeding / spotting — highest-weight term in Etonogestrel and Nexplanon; prominent in all four

✔️ Weight gain (gained, pound) — present in multiple topics per drug across all four corpora

✔️ Headache / migraine — consistent secondary complaint; particularly prominent in implant drugs

✔️ Acne — present in all four; strongest signal in Levonorgestrel

✔️ Decreased libido (sex drive) — present across all; standalone topic in Levonorgestrel only

✔️ Nausea — consistent secondary term; most prominent in oral formulations

These class-level signals broadly align with known hormonal contraceptive adverse effects listed in prescribing information. Their independent emergence across four separately processed corpora is strong evidence that the NMF method is detecting real clinical signal.

Drug-Specific Pharmacovigilance Signals

Drug	Specific Signal Not Prominent in Others
Etonogestrel / Nexplanon	***Contraceptive failure anxiety as a standalone topic. Patients misinterpreting spotting as pregnancy risk despite >99% efficacy. This psychosocial signal is absent from formal adverse event reporting frameworks.***
Ethinyl estradiol / norethindrone	***Ovarian cyst co-occurring within the psychological complaint cluster. Concurrent gynaecological and psychological adverse events not typically linked in prescribing communication.***
Levonorgestrel	***Libido suppression as a primary, standalone adverse event topic rather than secondary symptom. Patient explicit attribution to estrogen is a distinct patient-communication signal.***
Norethindrone (within combined)	***Resignation and tolerance language (worth, hoping, improved) clustered with metabolic complaints, suggesting passive acceptance of adverse effects rather than active management.***

6.3 Cross-Branch Synthesis

The two branches complement each other both analytically and in terms of what they demonstrate about text mining as a methodology:

Dimension	Insight
Convergent findings	***The supervised branch confirms that patient language robustly encodes drug sentiment. The unsupervised branch shows that the same language also encodes specific adverse event themes. Together, they confirm review text as a high-information source for pharmacovigilance.***
Methodological demonstration	***The supervised branch shows Orange can handle a full classification pipeline from raw text to evaluated model. The unsupervised branch shows Orange can extract clinically interpretable themes without any prior labelling. Both skills are demonstrated in a single project.***
Orange as a platform	***Both branches highlight the same core Orange strengths: Preprocess Text as a unified normalisation interface, Bag of Words***

	handling TF-IDF computation in the background, Test and Score for simultaneous multi-model evaluation, and Word Cloud for immediate interpretable feedback at every pipeline stage.
Practical value	***A pharmaceutical company could deploy the supervised branch as a real-time sentiment classifier for incoming reviews and the unsupervised branch as a periodic adverse event surveillance tool both running on the same text data pipeline.***

PHASE 6	Reflection — The Three Ws

7. Reflection: What Went Well, What Went Wrong, What I Would Do Differently

7.1 What Went Well

- **Orange's Preprocess Text widget** handled the entire text normalisation pipeline in one configurable interface, making iterative experimentation fast. Changes to one step propagated automatically to all downstream widgets without re-running the pipeline manually.
- **TF-IDF computation in the background** via Bag of Words removed significant implementation burden. The analyst focused on the analytical decision (which weighting scheme) rather than the computation appropriate for a text mining course context.
- **Test and Score's simultaneous multi-model evaluation** was the most time-saving Orange feature in the project. Four classifiers compared under identical conditions in a single widget with a side-by-side metrics table with no manual result tracking required.
- **Word Cloud as a preprocessing diagnostic** was invaluable. Reviewing the cloud after each pipeline run immediately revealed which stopwords were missing, which tokens were noise, and whether the vocabulary was clinically meaningful without writing any inspection code.
- **Using genuine new reviews for class augmentation** rather than duplicating existing ones was the right decision. It introduced real linguistic diversity to the negative class rather than simply inflating existing patterns.
- **NMF for topic modelling** produced cleaner and more interpretable adverse event themes than LDA for these short review texts. The decision to switch methods was validated by the clarity and clinical relevance of the final topic outputs.
- **Per-drug preprocessing tuning** in the unsupervised branch was necessary and effective. Applying a single shared configuration across corpora of very different sizes would have over-filtered small corpora and under-filtered large ones.

7.2 What Went Wrong

Issue 1 — Stopwords Set to Finnish (Not English)

The Preprocess Text filtering defaulted to Finnish. English stopwords (the, is, my, was, a) passed through unfiltered for several pipeline iterations, inflating the feature space with non-informative tokens. This was only detected through Word Cloud inspection. Fix: Changed to English. Prevention: Always verify the stopword language setting immediately when opening the widget.

Issue 2 — N-gram Range Excluded Unigrams (Set to 2–3)

The N-gram range was set to start at 2, which completely excluded single words. The most powerful sentiment features (terrible, effective, horrible, helped, pain, nausea) were all stripped from the feature set. Models were receiving only sparse bigrams and trigrams. Fix: Changed to 1–2. Prevention: Always check N-gram minimum is 1.

Issue 3 — Sentiment Not Recognised as Target Variable

The most persistent debugging challenge. Sentiment was typed as String in the original CSV. Even after fixing to Categorical, Orange's text pipeline (Corpus → Preprocess Text → Bag of Words) stripped the target designation. Test and Score returned 'Train data input requires a target variable' across multiple attempted fixes. Final resolution: Select Columns widget placed after Bag of Words explicitly re-assigned sentiment to the Target slot. Prevention: Add Select Columns as a standard step in any Orange text classification pipeline.

Issue 4 — Data Leakage via Rating Column

Perfect scores (AUC = 1.000 for three models) on the first evaluation run. Root cause: the rating column, from which sentiment was directly derived, was still present in the feature set. Models learned the rating-to-label mapping trivially. Fix: Rating moved to Ignored in Select Columns. Prevention: Review every column in Select Columns before connecting to Test and Score and explicitly move any column that could encode the target to Ignored.

Issue 5 — Class Imbalance (72/27 Original Split)

Initial CA scores (~79%) appeared reasonable but MCC scores (~0.38–0.44) revealed the model was biased toward predicting Positive. A model predicting Positive for everything would already achieve 72% CA. Fix: Class augmentation from supplementary dataset achieved 54/45 balance. MCC adopted as the primary metric. Prevention: Always check class distribution with Distributions widget before modelling.

Issue 6 — Learner Connections Mapped to Wrong Ports

All four classifiers connected to Test and Score via Predictions → Data instead of Learner → Learner. Orange showed a red error icon on Test and Score. Fix: All connections deleted and re-established with correct port mapping via the connection dialog. Prevention: When connecting to Test and Score, always use the dialog to explicitly select Learner → Learner for model widgets.

Issue 7 — LDA Topic Coherence = nan

LDA returned nan coherence and near-identical topics across all topic slots for every drug corpus tested. Multiple configurations were attempted before the method was switched to NMF. Significant time was lost. Prevention: For short, informal review texts, start with NMF. LDA is better suited to longer, more formal documents (news articles, academic papers).

Issue 8 — HTML Artifacts in Text (wa, ha, ve tokens)

Reviews encoded apostrophes as '. The Regexp tokeniser \w+ split these into meaningless fragments which flooded topic outputs. Prevention: Always use Word Punctuation tokeniser for user-generated web content. Inspect preprocessing output via Word Cloud before connecting Topic Modelling.

7.3 What I Would Do Differently

3. **Verify column types at the source immediately after CSV loading.** A 2-minute check of sentiment = Categorical in CSV File Import would have prevented the longest debugging chain in the project.
4. **Run a Distributions widget on the target variable before any modelling.** Detecting the 72/27 imbalance early would have prompted augmentation before building the full pipeline, not after.
5. **Check N-gram range and stopword language first when opening Preprocess Text.** These two settings have the largest impact on feature quality and both have non-obvious defaults that degrade performance.
6. **Add Select Columns as a standard pipeline step from the start** rather than discovering its necessity through repeated target variable errors.
7. **Start with NMF for topic modelling on review data.** Do not attempt LDA first on short informal texts - the theoretical default is not always the practical best choice.
8. **Connect a Data Table widget at each major pipeline stage during initial setup.** Verifying data at CSV import, after Select Rows, after Corpus, and after Bag of Words would have caught most configuration errors before they propagated downstream.
9. **Build and load the custom stopword file before running the first Word Cloud,** not after. Reviewing known medical domain noise terms in advance reduces the number of iterative refinement cycles needed.

FINAL	Conclusion

8. Conclusion

This project demonstrates the practical application of text mining to a real-world healthcare dataset using Orange Data Mining, following the CRISP-DM process framework across two analytically distinct branches.

The supervised branch confirms that patient review text carries sufficient linguistic signal to classify drug experiences as positive or negative with meaningful predictive accuracy. Review language is not random, it encodes sentiment in patterns that machine learning classifiers can learn and generalise. This has direct implications for automated review monitoring systems in pharmaceutical and healthcare contexts.

The unsupervised branch demonstrates that the same text also encodes structured adverse event information that can be extracted without any labels. NMF topic modelling revealed seven class-level adverse events consistent across all four Birth Control drugs, and four drug-specific signals that extend beyond formal prescribing documentation. These findings illustrate the potential of patient review mining as a complementary pharmacovigilance tool, particularly for therapeutic areas where formal adverse event reporting is incomplete.

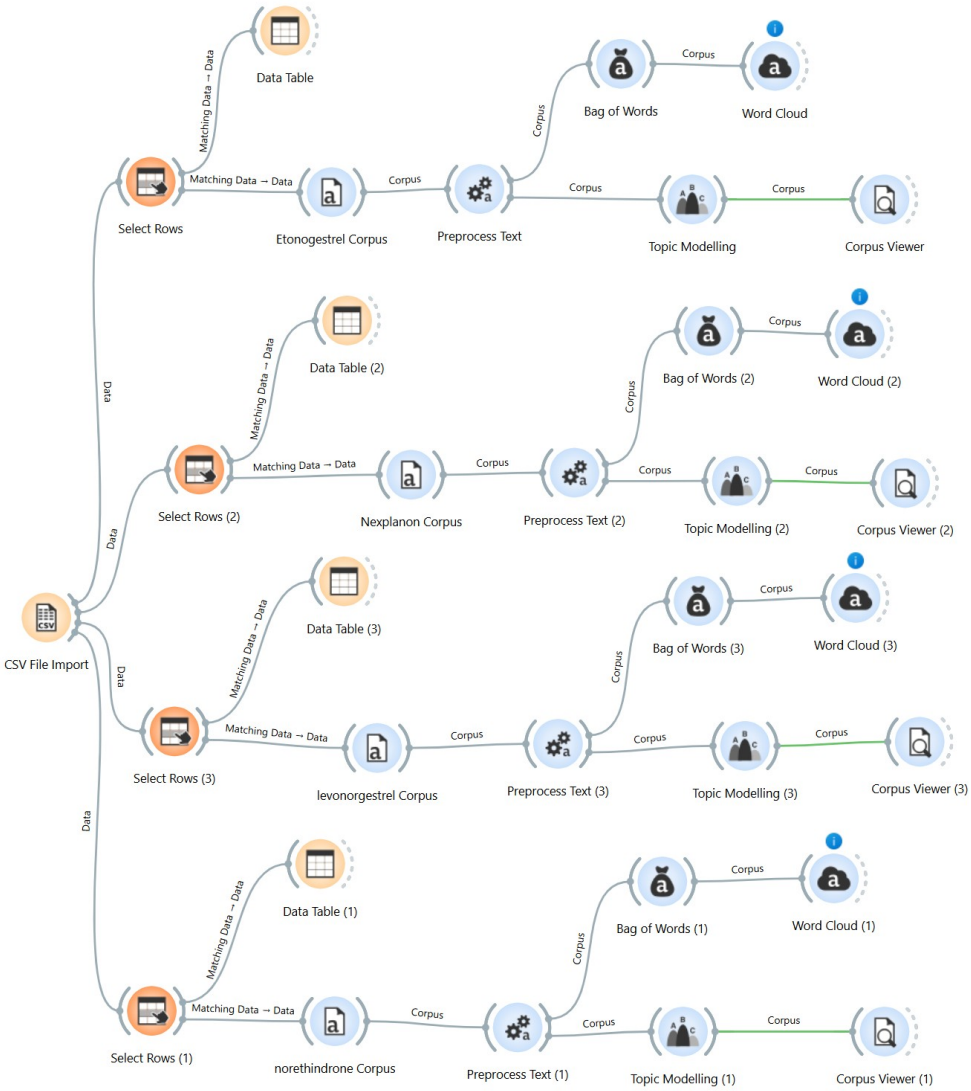
Together, the two branches make a coherent argument: patient review text is a high-information source that can both predict patient experience and reveal drug safety patterns. Orange provides an accessible, visual, and methodologically sound platform for conducting this type of analysis without requiring custom code, making text mining practical for analysts in clinical, regulatory, and pharmaceutical settings who do not have software engineering backgrounds.

Key Takeaways
1. Text alone predicts sentiment reliably (AUC > 0.79 at baseline; Final Model(Logistic Regression > 0.89 after full optimisation)
2. NMF is more suitable than LDA for short patient review texts
3. Preprocessing quality determines model quality. More time was spent on data preparation than modelling
4. Orange's visual pipeline makes iterative refinement fast and transparent
5. Class imbalance must be detected and addressed before trusting any performance metric
6. Patient reviews contain pharmacovigilance signals that formal reporting systems miss, particularly for psychological and psychosocial adverse events

Appendix	Orange Workflows

9. Workflows

9.1 Unsupervised Branch



9.2 Supervised Branch

